Answer Key & Teacher Notes

Session 1 — Volume of Rectangular Prisms · Session 2 — Missing Fractional Edges

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Volume of Rectangular Prisms

TALK PROMPTS

- How is counting layers of cubes the same as multiplying length × width × height?
- How many base layers stack up inside the locker? How do you know?
- Why is volume written in cubic units and not square units?

PARTNER TASK

Two Cartons. Each partner finds one carton's volume (18 × 5 × 11 = 990 cm³ and 15 × 11 × 5 = 825 cm³). Then explain which holds more and why the shared edges 11 and 5 make the comparison quick.

WATCH FOR

- Adding the three edges instead of multiplying them.
- Multiplying only two edges (finding base area) and calling it the volume.
- Writing square units (cm²) for a volume.

SESSION 2 · TEACHER NOTES

Missing Fractional Edges

TALK PROMPTS

- Why does dividing by the base area give back the missing edge?
- What happens to the missing edge if the volume stays the same but the base gets bigger?

PARTNER TASK

Two Unknowns. A prism has a volume of 270 cm³. Partner A finds the height when it is 9 cm by 5 cm (6 cm); Partner B finds the width when it is 6 cm long and 5 cm tall (9 cm). Check each other by multiplying all three edges.

WATCH FOR

- Dividing the volume by only one known edge.
- Treating 1 1/2 as 1 (or as 1.2) when multiplying.
- Writing a missing edge in cubic units (cm³) instead of length units.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.2	1	1	1	1	1	2	2	3	2	1	1
S1 · Form B	6.G.A.2	1	1	1	1	1	2	2	3	2	1	1
S2 · Form A	6.G.A.2	1	2	1	1	1	2	2	3	2	1	1
S2 · Form B	6.G.A.2	1	2	1	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

volume: The amount of **space** inside a solid.

unit cube: A cube with every edge **1** unit long.

base area: The area of the **bottom** face of a prism.

cubic units: Units used to measure **volume**.

- One layer of cubes holds length × **width** cubes.
- The height tells how many **layers** stack up.
- V = l × w × h, or V = **B** × h.

Example: One layer: 6 × 3 = **18** cubes. The height is 4, so there are **4** layers. Multiply: 18 × 4 = **72**. Answer: V = **72** cm³

YOU TRY

1. V = 5 × 2 × 3 = 30 in³

2. a. 60 m³ b. 25 × 4 = 100 ft³

EXIT TICKET

1. a. 42 cm³ b. 150 in³

2. V = 4 × 2 × 3 = 24 ft³

3. **C** (120 m³)

PRACTICE FORM A

1. a. 8 × 6 × 9 = 432 cm³ b. 7 × 4 × 3 = 84 ft³ c. 9 × 6 × 2 = 108 ft³ d. 42 × 7 = 294 in³

2. a. 5 × 4 = 20 cubes b. 3 layers c. 20 × 3 = 60 ft³

3. a. B = 16 × 11 = 176 in² b. V = 176 × 5 = 880 in³

4. B = 12 × 15 = 180 in²; V = 180 × 48 = 8,640 in³

5. V = 26 × 11 × 2 = 572 ft³

6. No. Ian multiplied only two edges (that is the base area). V = 17 × 13 × 5 = 1,105 cm³.

7. A: 990 cm³; B: 825 cm³. Carton A holds 165 cm³ more.

8. a. 60 × 30 × 40 = 72,000 cm³ b. 72,000 ÷ 1,000 = 72 L c. 60 × 30 × 35 = 63,000 cm³ = 63 L

9. Tomás added the edges. Volume multiplies them: V = 15 × 11 × 4 = 660 ft³.

10. **C** (60 cm³)

11. **C** (cubic units)

Self-check word **CAPACITY**

PRACTICE FORM B

1. a. 9 × 5 × 6 = 270 cm³ b. 8 × 3 × 2 = 48 ft³ c. 10 × 7 × 2 = 140 ft³ d. 36 × 8 = 288 in³

2. a. 6 × 3 = 18 cubes b. 4 layers c. 18 × 4 = 72 ft³

3. a. B = 18 × 12 = 216 in² b. V = 216 × 6 = 1,296 in³

4. B = 14 × 12 = 168 in²; V = 168 × 50 = 8,400 in³

5. V = 24 × 12 × 2 = 576 ft³

6. No. Rosa multiplied only two edges (that is the base area). V = 19 × 11 × 4 = 836 cm³.

7. A: 960 cm³; B: 840 cm³. Carton A holds 120 cm³ more.

8. a. 50 × 30 × 40 = 60,000 cm³ b. 60,000 ÷ 1,000 = 60 L c. 50 × 30 × 30 = 45,000 cm³ = 45 L

9. Malik added the edges. Volume multiplies them: V = 12 × 9 × 5 = 540 ft³.

10. **C** (60 m³)

11. **A** (base area)

Self-check word **CONTAINS**

SESSION 2 · ANSWERS

GUIDED NOTES

edge: A **segment** where two faces of a solid meet.

base area: The **product** of the length and width of the base.

inverse operations: Operations that **undo** each other.

- V = l × w × h works with **fractions** too.
- To find a missing edge, **divide** the volume by the product of the other two edges.
- Check: the three edges should **multiply** to the volume.

Example: Known edges: 6 × 4 = **24** cm². Divide: 72 ÷ 24 = **3**. Check: 6 × 4 × 3 = **72**. Answer: height = **3** cm

YOU TRY

1. V = 8 × 1/2 × 3 = 12 in³

2. a. 60 ÷ 12 = 5 m b. 9 ft³

EXIT TICKET

1. a. 15 cm³ b. 84 ÷ 12 = 7 in

2. 10 × 6 = 60; 180 ÷ 60 = 3 cm

3. **A** (6 ft³)

PRACTICE FORM A

1. a. 3/4 ft³ b. 77 in³ c. 336 ÷ 48 = 7 in d. 96 ÷ 12 = 8 in

2. a. 9 × 10 = 90 in² b. 990 ÷ 90 = 11 in c. Yes; 11 in is less than 13 in.

3. B = 13 × 10 = 130 cm²; h = 1,170 ÷ 130 = 9 cm

4. 20 × 16 × 3/4 = 320 × 3/4 = 240 in³

5. 20 × 12 × 1 1/2 = 240 × 1 1/2 = 360 in³

6. V = 22 × 9 × 1/2 = 99 ft³; cost 99 × 5 = $495

7. 30 × 18 = 540; 5,400 ÷ 540 = 10 in

8. a. 3/4 × 1/2 × 2 = 3/4 m³ b. 3/4 × 18 = $13.50 c. 1 1/2 m³; doubling the height doubles the volume.

9. Hana divided by only one edge. Divide by the product of both known edges: 11 × 9 = 99, and 1,980 ÷ 99 = 20 cm. Check: 20 × 11 × 9 = 1,980.

10. **A** (6 cm)

11. **A** (1 m³)

Self-check word **FRACTION**

PRACTICE FORM B

1. a. 1 1/2 m³ b. 54 in³ c. 240 ÷ 30 = 8 in d. 105 ÷ 15 = 7 in

2. a. 8 × 9 = 72 in² b. 1,080 ÷ 72 = 15 in c. No; 15 in is more than 13 in.

3. B = 12 × 7 = 84 cm²; h = 1,008 ÷ 84 = 12 cm

4. 24 × 12 × 3/4 = 288 × 3/4 = 216 in³

5. 18 × 10 × 1 1/2 = 180 × 1 1/2 = 270 in³

6. V = 20 × 8 × 1/2 = 80 ft³; cost 80 × 6 = $480

7. 25 × 20 = 500; 4,500 ÷ 500 = 9 in

8. a. 3/4 × 1/2 × 4 = 1 1/2 m³ b. 1 1/2 × 20 = $30 c. 3 m³; doubling the height doubles the volume.

9. Jonah divided by only one edge. Divide by the product of both known edges: 12 × 10 = 120, and 2,520 ÷ 120 = 21 cm. Check: 21 × 12 × 10 = 2,520.

10. **A** (8 in)

11. **A** (2 ft³)

Self-check word **MULTIPLY**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · VOLUME OF RECTANGULAR PRISMS

1. 4 × 3 × 2 = 24 cm³

2. a. 80 in³ b. 400 cm³ c. 216 ft³ d. 315 m³ e. 180 m³ f. 288 in³

3. V = 6 × 4 × 1 = 24 ft³; 24 ÷ 2 = 12 bags

4. B = 20 × 10 = 200 in²; h = 3,000 ÷ 200 = 15 in

5. No. A = 6 × 4 × 2 = 48 in³ and B = 3 × 4 × 4 = 48 in³. They hold the same amount.

SHEET 2 · MISSING FRACTIONAL EDGES

1. 4 × 3/2 × 5/2 = 60/4 = 15 in³

2. a. 24 ft³ b. 28 m³ c. 5/2 × 12 = 30 cm³ d. 9/4 × 4 = 9 in³ e. 4 × ½ × 5 = 10 ft³ f. 6 × 3/2 × 3 = 27 cm³

3. 3 × 1 ½ = 4 ½ ft²; 9 ÷ 4 ½ = 2 ft

4. a. 4 × 2 × 3 = 24 cubes b. 24 × 1/8 = 3 in³ (check: 2 × 1 × 1 ½ = 3)

5. **D** (15 in³)

6. **A** (4 ft)

October 2026 conceptual additions — teacher key

Session 1 · Volume of Rectangular Prisms

One layer has 4 × 3 = 12 cubes, and two layers make 24 cubes, or 24 cm³. The base area is 12 cm²; stacking through height produces volume.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: One layer has 4 × 3 = 12 cubes, and two layers make 24 cubes, or 24 cm³. The base area is 12 cm²; stacking through height produces volume.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Missing Fractional Edges

2 × 2 × 1 = 4 small cubes. Each has volume $(1/2)^3 = 1/8$ cm³; total $4/8 = 1/2$ cm³, agreeing with the edge product.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 2 × 2 × 1 = 4 small cubes. Each has volume $(1/2)^3 = 1/8$ cm³; total $4/8 = 1/2$ cm³, agreeing with the edge product.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.